Amine Benmansour

Senior Java engineer and tech lead · banking systems, cards and payments

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PROFILE

Senior Java engineer and tech lead with 14 years of experience, 11 of them inside the systems of six French banking institutions, from private banking to an online bank. I specialise in cards and mobile payments (Apple Pay and Google Pay tokenisation, the balance check with the card processor and Visa), fraud and disputes, and real-time flows with Kafka. On my banking assignments I owned application and data-flow architecture decisions, alone or with the lead architect: service boundaries, Kafka events, core-banking and mainframe integration, error recovery without duplicates. I bring AI into teams' daily work where it saves time, for instance to analyse production incidents. For confidentiality, clients are not named here; named references on request.

SKILLS

Card payments	issuing and card lifecycle (activation, blocking, PIN), Apple Pay / Google Pay tokenisation, balance check and Visa response-time constraints, fraud, disputes
Architecture	application and data-flow architecture, microservices, event-driven architecture, DDD, hexagonal architecture, Event Sourcing / CQRS, sagas and compensation, idempotency, REST / OpenAPI, ADRs
Flows and data	Kafka (Avro contracts, Schema Registry, retry, dead-letter), IBM MQ, RabbitMQ, ActiveMQ, Spring Batch, CFT, PostgreSQL, Oracle, MongoDB, SQL Server, Flyway
Integration	core banking, card processor, IBM mainframe (CICS / CTG), OAuth 2.0, SAML
Platform	Kubernetes (GKE, OpenShift), Google Cloud, Helm, Kustomize, ArgoCD, GitLab CI/CD, Jenkins, SonarQube
Operations	Dynatrace, Grafana, Elasticsearch / Kibana, Datadog, Gatling load testing, production incident management
Development	Java 8 to 21, Spring Boot 2 and 3, GraalVM, TDD / BDD (Cucumber), JUnit, Testcontainers
Regulation	PCI DSS, DORA, AML (KYC), ACPR requirements
Applied AI	coding agents, MCP servers (code repository, tickets, documentation, logs, image registry), GitHub Copilot, Claude Code

EXPERIENCE

Major French retail bank

December 2025 – present

Senior Java engineer, application and data-flow architecture

- Customer fraud-confirmation journey (card, transfer, cheque): design and development of the Java 21 / Spring Boot 3 microservices, the Kafka flows with Avro contracts, the REST APIs and the mainframe exchanges (CICS / CTG), and later enhancements.
- Error recovery without duplicate effects: Kafka retry strategy, replayable dead-letter topic and a step-tracking table. Each successful step is recorded and never replayed, so a replay never blocks a card twice; alerting on failures.
- Resilience pattern documented and presented to other teams for Kafka consumers that chain several external calls: orchestration, PostgreSQL step-tracking table, dead-letter and monitoring.
- Applied AI: incident-analysis agent run on demand by a business analyst or a developer, with the AI tool approved by the client (logs, tickets, monitoring and deployed images, through MCP servers); presented to other teams. Agent-assisted CVE remediation.
- Secure file-transfer component: Spring Batch, encryption, CFT.

Java 21, Spring Boot 3, Kafka, Avro, REST, CICS / CTG, PostgreSQL, Spring Batch, CFT, Kubernetes, ArgoCD, GitLab CI, Dynatrace

French online bank

January 2023 – September 2025

Tech lead, card and tokenisation platform

- Application architecture and development of the Java 21 / Spring Boot 3 microservices for the card lifecycle (issuing, activation, blocking, PIN) and Apple Pay / Google Pay tokenisation, in hexagonal architecture.
- Tokenisation performance: network latency traced end to end in Dynatrace, microservices merged to remove intermediate calls, request-reply over Kafka replaced by direct HTTP calls, PostgreSQL indexes reworked.
- Balance check between the bank, the card processor and Visa: investigated the bank’s response times and the use of stand-in authorisation.
- Data-flow architecture: Kafka event-driven design, and the trade-off between synchronous calls and Kafka to meet Visa's response times; REST / OpenAPI contracts with the card processor and the core banking system.

- Latency between the public cloud and on-premises: interconnect gateway and direct DNS resolution checked with the cloud network team.
- Migration from Spring Boot 2 to 3 and GraalVM Native Image; GitLab and ArgoCD CI/CD on Kubernetes, SonarQube, CVE tracking, Datadog and Dynatrace observability; Clean Code and TDD / BDD spread through code review.

  Java 21, Spring Boot 3, Kafka Confluent and Schema Registry, REST / OpenAPI, Kubernetes in the public cloud and on-premises, ArgoCD, GitLab CI/CD, PostgreSQL, GraalVM, Datadog, Dynatrace

Tier-1 French universal bank July 2021 – December 2022

Team lead and Java developer, retail banking document management APIs

- Architecture of the Spring Boot APIs in front of document management and of the Spring Batch jobs, within a SAFe train.
- Gatling load tests: bottlenecks found and fixed (saturated connection pool, slow queries), response times reduced; deployment automation, code review, onboarding of newcomers.

  Java 8, Spring Boot 2, Spring Batch, MongoDB, S3-compatible object storage, Gatling, Jenkins

IT division of a major French banking group October 2019 – July 2021

Tech lead alongside the architect, group private cloud portal

- Event Sourcing / CQRS application architecture and the portal's state machine; stateless Spring Boot microservices built with TDD.
- Automated virtual-machine provisioning flows and multi-provider management over RabbitMQ; attribute-based access control (ABAC), cost chargeback.

  Java 8, Spring Boot, Event Sourcing / CQRS, RabbitMQ, MongoDB, PostgreSQL, Jenkins

Specialised financial institution February 2018 – September 2019

Development team lead, customer portal for financing products

- Application architecture of the new customer portal in Clean Architecture and of the REST exchanges; technical leadership and code review; REST services with TDD, Jenkins CI, OpenShift deployment.

  Java 8 / 10, Spring Boot 2, JAX-RS, JPA, Angular, Oracle, Jenkins, OpenShift

Private bank of a major banking group September 2015 – February 2018

Java developer and team lead, KYC, customer risk rating and identity federation

- Private-bank applications in hexagonal architecture (customer risk rating under AML and ACPR requirements, KYC completeness, mobile customer area); identity federation with a SAML identity provider (OpenSAML); life-insurance application services.

  Java 7 / 8, Spring / Spring Boot, JBoss, SQL Server, OpenSAML, Jenkins

EARLIER EXPERIENCE

Sopra Steria, for a major Paris-region public transport operator (February 2012 – August 2015): Java / JEE developer. Real-time passenger information flows (OSGi / Karaf, Apache Camel, CXF, ActiveMQ), traffic control interface, geographic reference data (Hibernate, PostgreSQL).

PERSONAL PROJECTS

IAMINE Compta (2026): double-entry accounting on top of a banking API; PostgreSQL ledger whose balance is enforced by the database, chained immutable entries, bank ingestion published to Kafka with one idempotency key per transaction (Quarkus / GraalVM). Replaying a full fiscal year gives the same income statement as the accounting firm's.

Liam Medical (2025 – 2026): medical practice software designed alone and delivered to a practice in March 2026 (Java 21, Spring Boot 3, PostgreSQL); the AI assistance runs locally, so a patient's record never goes to an online service.

EDUCATION AND LANGUAGES

Master's degree in business IT (MIAGE), Université Nice Sophia Antipolis (2012). French (native), professional English, Arabic.